Hypothesis Testing with E-Values

Chapter 9: FDR Control using Compound E-Values

Based on Ramdas & Wang

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Why we need something beyond single-hypothesis testing I

Intuition. So far in this book we tested *one* hypothesis at a time with a single e-value. In practice, scientists rarely test just one thing: a genomics lab tests thousands of genes at once, a trading firm evaluates hundreds of strategies at once, a quality-control team checks dozens of production lines at once. If you test K hypotheses at level α each, purely by chance you expect around $K\alpha$ false rejections even when *every* hypothesis is null. With $K = 1000$ and $\alpha = 0.05$, that is about 50 false alarms – an unacceptable number if you then trust every rejection at face value.

The question this chapter answers: if we have one e-value (or more generally a "compound" e-value) per hypothesis, how do we combine them into a single multiple-testing procedure that controls a sensible aggregate error notion, no matter how the hypotheses/tests are statistically entangled with one another?

Roadmap.

- 9.1: the basic object (compound e-values) and the main procedure (e-BH).
- 9.2: every valid procedure secretly *is* e-BH (a universality result).
- 9.3: how to build the best possible compound e-values in a stylized model.
- 9.4–9.6: three ways to squeeze more discoveries out of e-BH for free (adaptivity, randomization, closure).

Why we need something beyond single-hypothesis testing II

Setting up the scene. We observe data X (the entire dataset) drawn from some unknown $\mathbb{P}^\dagger$. We want to test K null hypotheses simultaneously:

$$H_k : \mathbb{P}^\dagger \in \mathcal{P}_k, \qquad k \in \mathcal{K} := \{1, \ldots, K\}.$$

Each $\mathcal{P}_k$ is a set of candidate distributions consistent with "hypothesis k is true." The (unknown) index set of *true* null hypotheses is

$$\mathcal{N}(\mathbb{P}) = \{k \in \mathcal{K} : \mathbb{P} \in \mathcal{P}_k\}, \qquad K_0 = |\mathcal{N}(\mathbb{P})|.$$

Nothing here assumes independence across k: the K tests can be built from the same dataset, from overlapping data, or from entirely dependent processes (e.g. correlated stock returns, or genes on the same biological pathway).

Notation you'll need I

Intuition first: the spam-filter analogy. Suppose a spam filter flags 100 incoming emails as spam. Some of those 100 really are spam (correct rejections of "this is not spam"); some are perfectly innocent emails that got flagged by mistake (false discoveries). The *false discovery rate* (FDR) asks: *on average, what fraction of the flagged emails are actually not spam?* If the filter always flags 100 emails and, on average, 5 of them are legitimate, the FDR is 5%. This is exactly the right way to think about the FDR in multiple hypothesis testing: "discoveries" $=$ rejected hypotheses, and we want only a small fraction of them to be false alarms.

Notation you'll need II

False discovery proportion (FDP) and false discovery rate (FDR)

Let $\mathcal{D} \subseteq \mathcal{K}$ be the (random) set of hypotheses that a procedure rejects (the "discoveries"). Define

$$F_{\mathcal{D}} = |\mathcal{N}(\mathbb{P}) \cap \mathcal{D}| \quad \text{(number of \textit{false} discoveries)}, \qquad R_{\mathcal{D}} = |\mathcal{D}| \quad \text{(total discoveries)}.$$

$$\mathsf{FDP} = \frac{F_{\mathcal{D}}}{R_{\mathcal{D}} \vee 1}, \qquad \mathsf{FDR} = \mathbb{E}^{\mathbb{P}}[\mathsf{FDP}].$$

Here $F_{\mathcal{D}}$ counts true nulls that got rejected, $R_{\mathcal{D}}$ is how many hypotheses were rejected in total, and $a \vee b := \max(a, b)$ (so we never divide by zero: $0/0 := 0$). A procedure *controls FDR at level* α if FDR $\leq \alpha$ for every possible data-generating $\mathbb{P}$.

Notation you'll need III

Compound e-values (informal preview; formal definition in 9.1). In single testing, an e-value E for a null H_0 satisfies $\mathbb{E}^{\mathbb{P}}[E] \leq 1$ whenever H_0 is true. With K hypotheses, a natural generalization allows the K e-values $E_1, \ldots, E_K$ to *share* their total expected budget across only the currently-true nulls:

$$\sum_{k\,:\,\mathbb{P}\in\mathcal{P}_k} \mathbb{E}^{\mathbb{P}}[E_k] \leq K, \qquad \text{for every possible } \mathbb{P}.$$

This is looser than requiring each E_k individually to be a valid e-value for H_k – and this extra flexibility is the whole point of the chapter.

Notation you'll need IV

The classical Benjamini–Hochberg (BH) procedure (p-values). You may not have seen this before, so here is the core idea in full. Suppose $p_1, \ldots, p_K$ are p-values, one per hypothesis (small $p_k =$ evidence against H_k). Sort them from smallest to largest: $p_{(1)} \le p_{(2)} \le \cdots \le p_{(K)}$.

BH procedure at level α

Find the *largest* k such that the k-th smallest p-value clears a proportionally more lenient bar:

$$k^* = \max\left\{k \in \mathcal{K} : p_{(k)} \le \frac{k\alpha}{K}\right\}, \qquad (\max\emptyset := 0).$$

Reject the hypotheses corresponding to the k^* smallest p-values.

Why this works, intuitively: if you only ever compared $p_k \le \alpha$ (no correction), you would get about $K\alpha$ false rejections among K_0 true nulls – too many if K is large. BH instead lets the k-th smallest p-value clear a threshold that *grows* with k (namely $k\alpha/K$), which automatically adapts to how many hypotheses look interesting; the miracle (proved by Benjamini and Hochberg, 1995) is that this exact

Notation you'll need V

staircase controls FDR at level $(K_0/K)\alpha \leq \alpha$, provided the p-values are independent (or "PRDS", defined later).

Notation you'll need VI

From BH to e-BH. The e-BH procedure (Definition 9.8, next section) is the direct e-value analogue of BH: e-values point the opposite way from p-values (big $e =$ evidence against H_0, whereas small $p =$ evidence against H_0), so we sort e-values from *largest* to smallest and use a mirrored threshold condition. The huge payoff: e-BH controls FDR *regardless of the dependence structure* among the e-values – something the plain BH procedure cannot claim for p-values without extra assumptions (like independence or PRDS). This robustness to arbitrary dependence is the central theme of the whole chapter.

9.1.1 Compound e-variables: the formal definition I

Intuition. A single e-value must have expectation ≤ 1 under its own null. A *compound* e-value relaxes this: individual E_k's are allowed to have expectation larger than 1 (be "overconfident"), as long as the *total* expectation, summed only over hypotheses that happen to be truly null, never exceeds K (the total budget, one unit per hypothesis). This bookkeeping trick – pooling the error budget across hypotheses instead of hypothesis-by-hypothesis – is exactly what lets us build much more powerful and flexible procedures.

Definition 9.1: Compound e-variables

Fix nulls $(\mathcal{P}_1, \ldots, \mathcal{P}_K)$ and a set $\mathcal{P}$ of distributions. Random variables $E_1, \ldots, E_K \in [0, \infty]$ are *compound e-variables* for $(\mathcal{P}_1, \ldots, \mathcal{P}_K)$ under $\mathcal{P}$ if

$$\sum_{k\,:\,\mathbb{P} \in \mathcal{P}_k} \mathbb{E}^{\mathbb{P}}[E_k] \leq K \qquad \text{for all } \mathbb{P} \in \mathcal{P}.$$

They are *tight* if the supremum of the left-hand side over $\mathbb{P} \in \mathcal{P}$ equals K exactly (no wasted budget).

9.1.1 Compound e-variables: the formal definition II

Here the sum ranges only over currently-*true*-null indices k (those with $\mathbb{P} \in \mathcal{P}_k$); no restriction at all is placed on how $E_1, \ldots, E_K$ depend on each other. When $K = 1$ this reduces exactly to the ordinary definition of an e-variable.

9.1.1 Compound e-variables: the formal definition III

Example 9.2 (Weighted e-values). If each E_k is already a genuine e-variable for $\mathcal{P}_k$ (so $\mathbb{E}^{\mathbb{P}}[E_k] \leq 1$ whenever $\mathbb{P} \in \mathcal{P}_k$), then $E_1, \ldots, E_K$ are trivially compound e-variables. More generally, if $w_1, \ldots, w_K \geq 0$ are deterministic weights with $\sum_k w_k \leq K$, then $\tilde{E}_k := w_k E_k$ are also compound e-variables. Compound e-values are strictly more flexible than such weighted e-values, because weighting requires the stronger, hypothesis-by-hypothesis bound

$$\sum_{k \in \mathcal{K}} \sup_{\mathbb{P} \in \mathcal{P}_k} \mathbb{E}^{\mathbb{P}}[\tilde{E}_k] \leq K,$$

whereas compound e-values only need the pooled bound $\sup_{\mathbb{P}} \sum_{k \in \mathcal{N}(\mathbb{P})} \mathbb{E}^{\mathbb{P}}[E_k] \leq K$.

Example 9.3 (Multi-armed bandit / trader testing). Suppose K traders (or algorithms) are being evaluated. H_k: trader k has no genuine skill (their per-period return has conditional mean ≤ 1, i.e., no edge over a risk-free rate). Because trading strategies evolve over time and interact with a shared, evolving market, the realized performances $X_{k,1}, \ldots, X_{k,n}$ have complicated dependence both within a trader's own track record and across traders. P-values would be extremely hard to construct honestly here. But e-values are easy: $E_k = \prod_{t=1}^{n} X_{k,t}$ (compounded wealth from betting) is a valid e-value for H_k regardless of this dependence, because it is built from a sequential test-martingale. If traders start with different initial capital $c_1, \ldots, c_K$, then $E'_k/\bar{c}$ (where $\bar{c}$ is the average capital) gives compound e-variables, even though E'_k/c_k alone would be needed for each to be an individual e-variable.

9.1.1 Compound e-variables: the formal definition V

Examples 9.4–9.5 (two general recipes).

Example 9.4: Convex combinations

If $E_1^{(\ell)}, \ldots, E_K^{(\ell)}$ are compound e-variables for $\ell = 1, \ldots, L$, and $w_1, \ldots, w_L \geq 0$ are deterministic weights summing to 1, then $E_k := \sum_{\ell=1}^{L} w_\ell E_k^{(\ell)}$ are also compound e-variables. (Averaging across several candidate compound e-value constructions is always safe.)

Example 9.5: Adding follow-up evidence

Given compound e-variables $E_1, \ldots, E_K$, suppose we select a subset $\mathcal{S} \subseteq \mathcal{K}$ for further study, and collect new e-variables E'_j $(j \in \mathcal{S})$ that are valid *conditionally on* the first-stage data (i.e. $\mathbb{E}^{\mathbb{P}}[E'_j \mid E_j] \leq 1$). Setting $E'_j = 1$ for $j \notin \mathcal{S}$, the products $E_1 E'_1, \ldots, E_K E'_K$ are again compound e-variables. This licenses combining evidence from a pilot study and a follow-up study.

Intuition. Suppose you have one (compound) e-value per hypothesis: large e_k = strong evidence that H_k is false. The naive rule "reject if $e_k \geq 1/\alpha$" controls the error for each hypothesis *individually* (Markov's inequality), but doing this for all K hypotheses simultaneously lets false discoveries pile up, exactly like the naive p-value rule. The fix, mirroring BH: sort the e-values from *largest* to smallest, and find how far down the list you can go before the evidence, adjusted for how many things you are simultaneously rejecting, gets too weak.

Definition 9.8: The e-BH procedure

Let $e_1, \ldots, e_K$ be realized compound e-values, and let $e_{[1]} \geq e_{[2]} \geq \cdots \geq e_{[K]}$ denote them sorted from *largest to smallest* ($e_{[k]}$ is the k-th order statistic from the top). The e-BH procedure at level $\alpha \in (0,1)$ rejects the hypotheses with the largest k^* e-values, where

$$k^* := \max\left\{k \in \mathcal{K} : \frac{k\, e_{[k]}}{K} \geq \frac{1}{\alpha}\right\}, \qquad \max(\emptyset) := 0.$$

Here K is the total number of hypotheses, α is the target FDR level, and the condition $\frac{ke_{[k]}}{K} \geq \frac{1}{\alpha}$ says: "the k-th largest e-value, discounted by a factor k/K for how many rejections we are making, still clears

9.1.2 The e-BH procedure: intuition and definition II

the bar $1/\alpha$." We take the largest such k: rejecting more hypotheses is better (more discoveries), so we push k as far as the threshold condition allows.

9.1.2 The e-BH procedure: intuition and definition III

Step-by-step: how to run the e-BH procedure by hand.

1. **Sort** the K compound e-values from largest to smallest: $e_{[1]} \geq e_{[2]} \geq \cdots \geq e_{[K]}$.
2. **Scan down the sorted list**, and for each rank $k = 1, 2, \ldots, K$ check whether

$$k \cdot e_{[k]} \;\geq\; \frac{K}{\alpha}.$$

3. **Find** k^*, the *largest* rank k for which this check passes (if none pass, $k^* = 0$ and nothing is rejected).
4. **Reject** the k^* hypotheses with the largest e-values (i.e., $e_{[1]}, \ldots, e_{[k^*]}$).

9.1.2 The e-BH procedure: intuition and definition IV

An equivalent, threshold-based description (Proposition 9.12). Define, for any $t \geq 0$, $R(t) = |\{k : e_k \geq t\}| \vee 1$ (how many e-values are $\geq t$), and let

$$t_\alpha = \inf\{t \geq 0 : t\, R(t) \geq K/\alpha\}.$$

Then H_k is rejected by e-BH if and only if $e_k \geq t_\alpha$: e-BH is exactly "reject everything above a single data-dependent cutoff t_α," and this cutoff always takes one of the values $K/(\alpha k)$, $k \in \mathcal{K}$. This alternative view will be useful later (Section 9.1.3) when boosting e-values, and it clarifies why e-BH is really just a smart, self-calibrating threshold.

Running example: 10 coins, set up I

The scenario. You have $K = 10$ coins. For each coin k, the null hypothesis is H_k: "coin k is fair." Using the data from flipping each coin many times, a sequential betting scheme (as in earlier chapters) produces one e-value e_k per coin: the larger e_k, the stronger the evidence that coin k is biased. We will use *this same* numeric example throughout the rest of the chapter (Sections 9.1, 9.4, 9.5, 9.6) so you can track how each new refinement changes the verdict.

Coin k	1	2	3	4	5	6	7	8	9	10
e_k	0.6	12.0	0.4	34.0	0.8	90.0	0.5	200.0	24.0	0.7

Table: Our running example: $K = 10$ coins, one e-value each.

We will test all 10 hypotheses at once, at level $\alpha = 0.1$. Coins 1, 3, 5, 7, 10 look fair (e-values near or below 1); coins 4, 6, 8, 9 look suspicious (e-values far above 1); coin 2 is a border case (e-value moderately above 1).

Running example: 10 coins, set up II

Step 1: sort the e-values from largest to smallest.

rank k	1	2	3	4	5	6	7	8	9	10
$e_{[k]}$	200.0	90.0	34.0	24.0	12.0	0.8	0.7	0.6	0.5	0.4
coin	8	6	4	9	2	5	10	1	7	3

Step 2: compute the threshold $K/\alpha = 10/0.1 = 100$, and check $k \cdot e_{[k]} \geq 100$ at each rank:

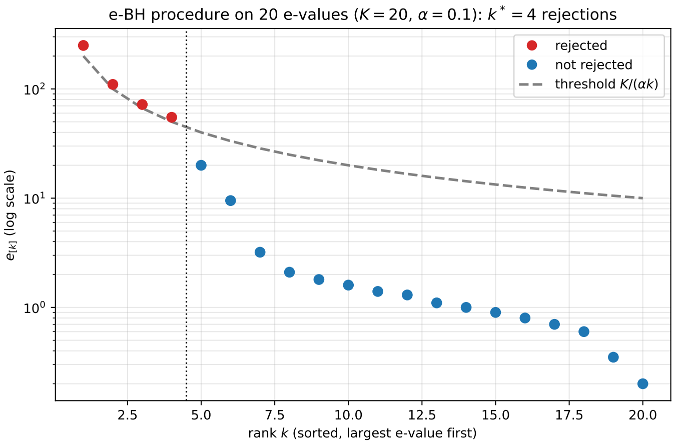
k	$e_{[k]}$	$k \cdot e_{[k]}$ vs. 100
1	200.0	$1 \times 200.0 = 200.0 \ \geq 100$ ✓
2	90.0	$2 \times 90.0 = 180.0 \ \geq 100$ ✓
3	34.0	$3 \times 34.0 = 102.0 \ \geq 100$ ✓ (just barely!)
4	24.0	$4 \times 24.0 = 96.0 \ < 100$ ×
5	12.0	$5 \times 12.0 = 60.0 \ < 100$ ×
6–10	≤ 0.8	even smaller: all fail

Running example: 10 coins, set up III

Step 3–4: find k^* and reject. The largest rank at which the check passes is $k^* = 3$ (rank 4 fails, at $96.0 < 100$, and nothing beyond it passes either). So e-BH **rejects the top 3**: coins 8, 6, 4 (e-values $200.0, 90.0, 34.0$).

Reading the near-miss. Notice how close rank 3 was: $3 \times 34.0 = 102.0$ barely clears 100, while rank 4 misses by only 4 points (96.0 vs. 100). Coin 9 (e-value 24.0) is the most interesting near-miss – Sections 9.4–9.6 will revisit exactly this coin and show how smarter (but still fully valid) procedures manage to reject it too.

Figure: the e-BH procedure on a toy set of e-values



Sorted e-values (largest first) vs. the e-BH threshold curve $K/(\alpha k)$, $K = 20, \alpha = 0.1$. Points above the dashed curve are rejected ($k^* = 4$).

Python: the e-BH procedure on our 10 coins

```python
import numpy as np

e = np.array([0.6, 12.0, 0.4, 34.0, 0.8,
              90.0, 0.5, 200.0, 24.0, 0.7])
coins = [f"Coin {i+1}" for i in range(10)]
K, alpha = len(e), 0.1

order = np.argsort(-e)          # sort descending
e_sorted = e[order]
k_idx = np.arange(1, K + 1)

# e-BH: k* = max{ k : k * e_[k] / K >= 1/alpha }
satisfies = (k_idx * e_sorted / K) >= (1 / alpha)
k_star = np.max(np.where(satisfies)[0]) + 1 if satisfies.any() else 0

rejected = [coins[i] for i in order[:k_star]]
print("Sorted e-values:", e_sorted)
print("k* =", k_star)
print("Rejected hypotheses:", rejected)

# --- output ---
# Sorted e-values: [200.   90.   34.   24.   12.    0.8  0.7  0.6  0.5  0.4]
# k* = 3
# Rejected hypotheses: ['Coin 8', 'Coin 6', 'Coin 4']
```

Why e-BH controls the FDR: self-consistency I

Intuition. We want to show that no matter how the compound e-values $E_1, \ldots, E_K$ depend on each other, e-BH's FDR never exceeds α. The key structural property of e-BH is that *every rejected hypothesis has a big enough e-value, relative to the total number of rejections*. We isolate this property and show it alone is enough for FDR control – this makes the proof apply to a whole family of related procedures at once (self-consistent ones), not just to e-BH.

Definition 9.9: Self-consistent e-testing procedures

An e-testing procedure $\mathcal{D}$ is *self-consistent at level* α if every rejected hypothesis H_k has realized e-value satisfying

$$e_k \ \geq \ \frac{K}{\alpha R_{\mathcal{D}}},$$

where $R_{\mathcal{D}} = |\mathcal{D}|$ is the total number of rejections. The e-BH procedure is a special case (in fact the *largest* self-consistent procedure: it dominates every other self-consistent procedure).

In words: if you reject $R_{\mathcal{D}}$ hypotheses total, each one better have an e-value of at least $K/(\alpha R_{\mathcal{D}})$ – the more you reject, the higher a bar each individual rejection must clear (since the same total error budget

Why e-BH controls the FDR: self-consistency II

K is being spread over more discoveries with a laxer per-rejection threshold as $R_{\mathcal{D}}$ grows, this exactly balances out, as the proof below shows).

Why e-BH controls the FDR: self-consistency III

Theorem 9.11: FDR control (and post-hoc FDR control)

Any self-consistent e-testing procedure at level α, including e-BH, has FDR at most α, for *any* $\mathbb{P}$ and *any* compound e-variables, regardless of their dependence structure. In fact, for any family $(\mathcal{D}_\alpha)_{\alpha \in (0,1)}$ of self-consistent procedures indexed by level,

$$\mathbb{E}^{\mathbb{P}}\left[\sup_{\alpha \in (0,1)} \frac{F_{\mathcal{D}_\alpha}}{\alpha R_{\mathcal{D}_\alpha}} \right] \leq 1,$$

a *post-hoc* guarantee valid even if α itself is chosen after looking at the data. If $E_1, \ldots, E_K$ happen to be ordinary e-variables (not just compound), the bound improves to K_0/K.

Why e-BH controls the FDR: self-consistency IV

Step-by-step proof walkthrough. Let $\mathbf{E} = (E_1, \dots, E_K)$ be arbitrary compound e-variables, and $\mathcal{D}_\alpha(\mathbf{E})$ the rejection set.

1. **Rewrite the FDP** using indicators: for the true-null set $\mathcal{N}$,

$$\frac{F_{\mathcal{D}_\alpha}}{\alpha R_{\mathcal{D}_\alpha}} = \frac{|\mathcal{D}_\alpha(\mathbf{E}) \cap \mathcal{N}|}{\alpha(R_{\mathcal{D}_\alpha} \vee 1)} = \sum_{k \in \mathcal{N}} \frac{\mathbb{1}\{k \in \mathcal{D}_\alpha(\mathbf{E})\}}{\alpha(R_{\mathcal{D}_\alpha} \vee 1)}.$$

2. **Bound each surviving term by self-consistency:** if $k \in \mathcal{D}_\alpha(\mathbf{E})$ (i.e., H_k was rejected), self-consistency says $E_k \geq K/(\alpha R_{\mathcal{D}_\alpha})$, i.e., $\frac{1}{\alpha(R_{\mathcal{D}_\alpha} \vee 1)} \leq \frac{E_k}{K}$. Substituting,

$$\frac{F_{\mathcal{D}_\alpha}}{\alpha R_{\mathcal{D}_\alpha}} \leq \sum_{k \in \mathcal{N}} \frac{\mathbb{1}\{k \in \mathcal{D}_\alpha(\mathbf{E})\}\, E_k}{K} \leq \sum_{k \in \mathcal{N}} \frac{E_k}{K}.$$

(The last step just drops the indicator, which can only increase the sum since $E_k \geq 0$.)

Why e-BH controls the FDR: self-consistency V

3. **Take expectations and invoke the compound e-value bound:** by Definition 9.1, $\sum_{k\in\mathcal{N}} \mathbb{E}^{\mathbb{P}}[E_k] \le K$, so
$$\mathbb{E}^{\mathbb{P}}\Big[\sup_{\alpha} \frac{F_{\mathcal{D}_\alpha}}{\alpha R_{\mathcal{D}_\alpha}}\Big] \le \mathbb{E}^{\mathbb{P}}\Big[\sum_{k\in\mathcal{N}} \frac{E_k}{K}\Big] \le 1.$$
4. **Drop the supremum** over α (keeping only a single fixed α) to recover the plain FDR bound: $\mathrm{FDR}_{\mathcal{D}_\alpha} \le \alpha$.

Notice that the whole argument never used independence, or any dependence assumption at all, among $E_1, \ldots, E_K$ – this is exactly why e-BH is dependence-robust.

9.1.3 Boosting e-values with distributional information I

Intuition. e-BH as defined only ever uses each e-value in a very coarse way: it only matters whether e_k clears certain grid thresholds $K/(\alpha k)$. If we know more about the null distribution of E_k, we can sometimes replace E_k by a strictly larger quantity $b_k E_k$ ($b_k \geq 1$, a "boosting factor") that *still* behaves like a valid e-value at the thresholds e-BH cares about – giving strictly more power for free.

The truncation function and boosting condition

Let $K/\mathcal{K} := \{K/k : k \in \mathcal{K}\}$ and define $T : [0, \infty] \to [0, K]$ by

$$T(x) = \frac{K}{\lceil K/x \rceil} \nVdash \{x \geq 1\}, \qquad T(\infty) = K,$$

i.e. T rounds x down to the nearest value in $K/\mathcal{K} \cup \{0\}$. A *boosting factor* $b_k \geq 1$ for hypothesis k must satisfy

$$\mathbb{E}^{\mathbb{P}}[T(\alpha b_k E_k)] \leq \alpha \qquad \text{for } \mathbb{P} \in \mathcal{P}_k. \tag{9.2}$$

The *boosted e-value* is $E'_k = b_k E_k$. Choosing $b_k = 1$ always works (since $T(x) \leq x$), but the goal is to find the *largest* valid b_k, which requires distributional knowledge of E_k under the null; with no such knowledge, $b_k = 1$ is forced.

9.1.3 Boosting e-values with distributional information III

Theorem 9.13

Applied to arbitrary nonnegative $E'_1, \dots, E'_K$, the e-BH procedure at level α satisfies

$$\mathbb{E}^{\mathbb{P}}\left[\frac{F_{\mathcal{D}}}{R_{\mathcal{D}}}\right] \le \frac{1}{K}\sum_{k\in\mathcal{N}} \mathbb{E}^{\mathbb{P}}[T(\alpha E'_k)].$$

In particular, if $E'_1, \dots, E'_K$ are the boosted e-values from (9.2), FDR is controlled at level $K_0\alpha/K \le \alpha$.

Intuition for why this works. Since e-BH only ever rejects at the grid thresholds $K/(\alpha k)$, replacing each e_k with $T(\alpha e_k)/\alpha$ (i.e., rounding it down to the nearest relevant grid level) produces the *exact same rejection set*. So it suffices that the *rounded-down* version of αE_k behaves like a bona fide e-value, which is precisely condition (9.2). We can boost individual e-values this way, or boost jointly across a whole compound e-value collection using boosting factors satisfying the analogous pooled condition $\sum_{k:\mathbb{P}\in\mathcal{P}_k} \mathbb{E}^{\mathbb{P}}[T(\alpha b_k E_k)] \le K\alpha$.

9.1.4 Connection between e-BH and BH I

Intuition. P-values and e-values are related by $P_k = 1/E_k$ (a valid e-value always gives a valid p-value this way). So it is natural to ask: is e-BH just BH in disguise? The answer is subtle: e-BH *is* literally BH applied to $1/E_1, \ldots, 1/E_K$, but the FDR guarantee for plain BH normally needs an extra dependence assumption (PRDS or independence) that e-BH does not need.

Definition 9.14: The BH and BHY procedures

With p-values $p_1 \leq \cdots \leq p_K$ sorted ascending, BH at level α rejects the k^* smallest, where $k^* = \max\{k : Kp_{(k)}/k \leq \alpha\}$. The *BHY* procedure (Benjamini–Yekutieli correction) applies BH to $\ell_K p_1, \ldots, \ell_K p_K$ instead, where $\ell_K = \sum_{k=1}^{K} 1/k \approx \log K$.

Definition 9.15: PRDS (positive regression dependence on a subset)

A set $A \subseteq \mathbb{R}^K$ is *increasing* if $\mathbf{x} \in A, \mathbf{y} \geq \mathbf{x} \Rightarrow \mathbf{y} \in A$. $(P_1, \ldots, P_K)$ satisfies PRDS if for every null index k and increasing set A, $x \mapsto \mathbb{P}\{(P_\ell)_{\ell \in \mathcal{K}} \in A \mid P_k \leq x\}$ is increasing on $[0, 1]$. (Informally: conditioning on a null p-value being small should not make other p-values look larger – a positive-dependence condition, often hard to verify in practice.)

9.1.4 Connection between e-BH and BH II

Classical facts. (i) BHY controls FDR at level $(K_0/K)\alpha$ under *arbitrary* dependence. (ii) Plain BH controls FDR at level $(K_0/K)\alpha$ *if* the p-values are independent or PRDS. The price for full dependence-robustness (BHY) is the extra factor $\ell_K \approx \log K$, which shrinks all p-values, and hence typically causes many fewer rejections.

Proposition 9.16 and the e-value viewpoint

Since $P_k = 1/E_k$ is always a p-value when E_k is an e-value, e-BH is *exactly* BH applied to $1/E_1, \ldots, 1/E_K$ – but e-BH's FDR guarantee (Theorem 9.11) needs no dependence assumption at all. Moreover, $f(p) = T(\alpha/(\ell_K p))/\alpha$ is a valid *calibrator* (turns a p-value into an e-value), and the BHY procedure equals e-BH applied to the calibrated e-values $f(P_1), \ldots, f(P_K)$.

Takeaway. e-BH quietly *contains* both BH (as a special "boosted e-values" instance, using knowledge of PRDS to construct the boost) and BHY (via the calibrator f) as special cases – this is a first hint of the universality theme of Section 9.2.

9.2: Every FDR procedure secretly is e-BH I

Intuition. This is one of the more surprising results in the chapter: *any* procedure that controls FDR – no matter how it was built, whether it uses e-values, p-values, or something else entirely – can be re-expressed as "run e-BH on some cleverly chosen compound e-values." This does not mean every procedure was secretly designed with e-values in mind; it means the e-BH template is rich enough to represent every valid FDR procedure after the fact.

Theorem 9.18: From FDR control to compound e-values

Let $\mathcal{D}$ be any procedure controlling FDR at level α. Let $V_k \in \{0,1\}$ indicate whether $\mathcal{D}$ rejects H_k (so $F_{\mathcal{D}} = \sum_{k\in\mathcal{N}} V_k$, $R_{\mathcal{D}} = \sum_{k\in\mathcal{K}} V_k$). Define

$$E_k = \frac{K}{\alpha} \cdot \frac{V_k}{R_{\mathcal{D}} \vee 1}. \tag{9.3}$$

Then $E_1, \ldots, E_K$ are compound e-variables for $(\mathcal{P}_1, \ldots, \mathcal{P}_K)$.

Why: $\sum_{k:\mathbb{P}\in\mathcal{P}_k} \mathbb{E}^{\mathbb{P}}[E_k] = \frac{K}{\alpha}\mathbb{E}^{\mathbb{P}}\left[\frac{F_{\mathcal{D}}}{R_{\mathcal{D}}\vee 1}\right] \leq \frac{K}{\alpha} \cdot \alpha = K$, using exactly the FDR guarantee of $\mathcal{D}$.

9.2: Every FDR procedure secretly is e-BH II

Theorem 9.19: Universality of e-BH

Let $\mathcal{D}$ be any FDR-controlling procedure at level α. Then there exist compound e-values such that e-BH applied to them makes *exactly* the same discoveries as $\mathcal{D}$. Further, if $\mathcal{D}$ is *admissible* (cannot be enlarged while still controlling FDR), these compound e-values can be chosen to be *tight*.

The proof takes the E_k's from (9.3) and, if they are not already tight (i.e. $K^* := \sup_{\mathbb{P}} \sum_{k:\mathbb{P}\in\mathcal{P}_k} \mathbb{E}^{\mathbb{P}}[E_k] < K$), rescales them by K/K^* to use up the full budget; admissibility then forces the rescaled e-BH procedure to coincide with $\mathcal{D}$ itself (otherwise it would strictly dominate $\mathcal{D}$, contradicting admissibility).

9.2: Every FDR procedure secretly is e-BH III

A practical payoff: combining and derandomizing procedures. Suppose L different multiple-testing procedures $\mathcal{D}_1, \ldots, \mathcal{D}_L$ are run (perhaps by different teams, using different methods, each controlling FDR at its own level α_ℓ), and we want to merge them into a single procedure. Recipe:

1. For each procedure ℓ, form compound e-variables $E_1^{(\ell)}, \ldots, E_K^{(\ell)}$ via Theorem 9.18 (this always exists).
2. Fix convex weights $w_1, \ldots, w_L \geq 0$, $\sum_\ell w_\ell = 1$ (possibly random, but independent of the e-values), and combine via Example 9.4: $E_k = \sum_{\ell=1}^{L} w_\ell E_k^{(\ell)}$.
3. Apply e-BH at level α to the combined $E_1, \ldots, E_K$.

This combined procedure controls FDR at level α – and the original levels α_ℓ do not even matter for the final guarantee! A key application is *derandomization*: if $\mathcal{D}_\ell$ depends on external randomness U_ℓ, averaging over many independent copies (as $L \to \infty$) removes the procedure's sensitivity to the random seed while still controlling FDR.

9.3: Motivation and Robbins' compound decision theory I

Intuition. So far we assumed compound e-values are handed to us. Section 9.3 asks: in a clean sequence model where each hypothesis k has its own data point X_k, what is the *best possible* compound e-value we could construct, if we are only allowed the simplest possible rule – apply the *same* function f to each X_k? This connects directly to a classical idea in statistics: Robbins' compound decision theory.

Definition 9.20: Simple separable e-variables

$E_1, \ldots, E_K$ are *separable* if $E_k = E_k(X_k)$ depends only on the k-th coordinate of the data. They are *simple separable* if, further, $E_k = f(X_k)$ for the *same* function f across all k.

Background (9.3.1): Robbins' compound decision problem. In the Gaussian sequence model, $X_k \sim N(\mu_k, 1)$ independently, $k \in \mathcal{K}$, with unknown means $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_K)$ fixed. We want estimators $\hat{\mu}_k$ minimizing the *compound* (averaged) risk $\frac{1}{K} \sum_k \mathbb{E}^{\boldsymbol{\mu}}[(\hat{\mu}_k - \mu_k)^2]$.

9.3: Motivation and Robbins' compound decision theory II

The fundamental theorem of compound decisions. Let $M = \frac{1}{K}\sum_k \delta_{\mu_k}$ be the empirical distribution of the true means (a single "prior"), and consider the associated univariate Bayes problem $\mu' \sim M,\ X' \mid \mu' \sim N(\mu', 1)$. For *any* fixed function $s : \mathbb{R} \to \mathbb{R}$,

$$\frac{1}{K}\sum_{k\in\mathcal{K}} \mathbb{E}^{\boldsymbol{\mu}}[(s(X_k) - \mu_k)^2] = \mathbb{E}^{\mathbb{P}_M}[(s(X') - \mu')^2].$$

So the best *simple separable* estimator s (same function applied to every X_k) is exactly the Bayes estimator under M: $s(x) = \mathbb{E}^{\mathbb{P}_M}[\mu' \mid X' = x]$. This gives an oracle benchmark: even though the oracle needs to know the true $\boldsymbol{\mu}$ (through M) to construct s, it is still constrained to use the *same* s for every coordinate – exactly like our compound e-value problem, where the oracle also "knows everything" but the estimator/e-value function itself must be simple and separable.

9.3.2: The log-optimal simple separable compound e-value. Suppose each null $\mathcal{P}_k = \{\mathbb{S} : \mathbb{S}_k = \mathbb{P}_k\}$ is a simple point null on the k-th marginal, with $d\nu$-density p_k, and suppose alternatives $\mathbb{Q}_k$ have densities q_k. We seek the simple separable function s maximizing the *compound expected log-wealth* (a natural notion of power for e-values):

$$\underset{s:\mathcal{Y}\to[0,\infty)}{\text{maximize}} \quad \frac{1}{K}\sum_{k\in\mathcal{K}} \mathbb{E}^{\mathbb{Q}_k}[\log(s(X_k))] \quad \text{s.t. } s(X_1),\dots,s(X_K) \text{ are compound e-variables.} \tag{9.8}$$

Theorem 9.21: Log-optimal simple separable compound e-values

The optimal solution to (9.8) is the likelihood ratio of the *mixtures*:

$$s(x) = \frac{\sum_{j\in\mathcal{K}} q_j(x)}{\sum_{j\in\mathcal{K}} p_j(x)}.$$

9.3: Motivation and Robbins' compound decision theory IV

Intuition for the proof: averaging both the objective and the compound e-value constraint over $k \in \mathcal{K}$ turns the whole compound problem into a single ordinary e-value optimization problem (of the type solved in Chapter 3) for testing the average null density $\bar{p} = \frac{1}{K}\sum_j p_j$ against the average alternative density $\bar{q} = \frac{1}{K}\sum_j q_j$; the log-optimal single e-value there is exactly the likelihood ratio $\bar{q}(x)/\bar{p}(x) = s(x)$.

Proposition 9.22: Tight simple separable compound e-values must look like this

If $E_1, \ldots, E_K$ are *tight* simple separable compound e-values, then there exist $d\nu$-densities $q_1, \ldots, q_K$ such that E_k has the form of Theorem 9.21 on the support of $\bar{p} = \frac{1}{K}\sum_k p_k$, and (without loss of generality) one may take $q_1 = \cdots = q_K$.

Why should we care? The optimal compound e-value $s(x) = \bar{q}(x)/\bar{p}(x)$ requires knowing $q_1, \ldots, q_K$ (the true alternatives) – information we rarely have exactly. But if K is large and all the alternatives share a common unknown structure (e.g., a shared nuisance parameter θ), we may be able to *learn* that shared structure by pooling data across all K hypotheses (empirical Bayes / compound decision theory), even though we could never learn it from a single hypothesis's data alone. This is the deep reason "compound" e-values, which allow such pooling, can outperform naive per-hypothesis e-values when K is large.

Python: log-optimal separable compound e-value (Gaussian model)

```
import numpy as np
rng = np.random.default_rng(1)

# H_k : X_k ~ N(0,1) (null), K=10. Believed alt. means (pooled/learned):
K = 10
mu = np.array([0,0,0,0,0,0, 2.0, 2.5, 3.0, 3.5])

def p_dens(x):    # common null density N(0,1)
    return np.exp(-0.5*x**2)/np.sqrt(2*np.pi)

def q_mix(x):     # mixture of alternative densities
    return np.mean([np.exp(-0.5*(x-m)**2)/np.sqrt(2*np.pi) for m in mu], axis=0)

def s(x):         # log-optimal simple separable e-value, Thm 9.21
    return q_mix(x) / p_dens(x)

# MC check: under the GLOBAL null, E[s(X_k)] <=1 each, sum <= K.
n_mc = 200_000
vals = [np.mean(s(rng.normal(0, 1, n_mc))) for _ in range(K)]
print("E[s(X_k)]:", np.round(vals, 3))
print("Sum:", round(sum(vals), 2), " (budget K =", K, ")")
# output: E[s(X_k)]: [0.985 1.061 0.99 0.991 1.009 0.994 1.057 0.99 1.002 0.983]
# output: Sum: 10.06  (budget K = 10)
```

9.4: Getting one more rejection almost for free I

Intuition. e-BH is sometimes needlessly conservative for small K. The fix here costs almost nothing: first spend a tiny sliver of the error budget testing the *global null* ("are all K hypotheses simultaneously true?"); if that global null is confidently rejected, we know *at least one* hypothesis is false, so we can afford to run e-BH slightly more aggressively (at a laxer level $\alpha' > \alpha$) on the rest, since one "guaranteed" true rejection is already banked.

The minimally adaptive e-BH procedure $\mathcal{D}_\alpha^F$

Fix an e-merging function $F : [0,\infty)^K \to [0,\infty)$ (Chapter 8) – e.g. the arithmetic mean $\mathbb{M}_K(e_1,\dots,e_K) = \frac{1}{K}\sum_k e_k$. Given realized e-values $e_1,\dots,e_K$ and level α:

tep 1. Test the global null $\bigcap_{k=1}^K \mathcal{P}_k$: reject it if $F(e_1,\dots,e_K) \geq 1/\alpha$.

tep 2. If the global null is *not* rejected, output $\mathcal{D}_\alpha^F = \emptyset$ (no discoveries).

tep 3. If the global null *is* rejected, run the ordinary e-BH procedure at the slightly larger level $\alpha' = K\alpha/(K-1)$, and output its rejection set.

9.4: Getting one more rejection almost for free II

Proposition 9.23

For any $\alpha \in (0,1)$, the improved procedure $\mathcal{D}_\alpha^F$ applied to arbitrary e-values has FDR at most α. Moreover, taking $F = \mathbb{M}_K$ (the arithmetic mean) makes $\mathcal{D}_\alpha^{\mathbb{M}_K}$ *dominate* plain e-BH: $\mathcal{D}_\alpha \subseteq \mathcal{D}_\alpha^{\mathbb{M}_K}$ always.

Why FDR is still controlled (sketch). If not all nulls are true $(K_0 < K)$, Theorem 9.11 applied at the larger level $\alpha' = K\alpha/(K-1)$ already gives FDR $\leq (K_0/K)\alpha' \leq \alpha$. If instead *all* hypotheses are truly null $(K_0 = K)$, the only way to make any discovery at all is to (falsely) reject the global null, which by Markov's inequality happens with probability at most α – so FDR $\leq \alpha$ in this case too. Either way, α is never exceeded.

9.4: Getting one more rejection almost for free III

Running example, continued: minimally adaptive e-BH on our 10 coins. Step 1: test the global null. Using $F = \mathbb{M}_K$ (arithmetic mean):

$$\bar{e} = \frac{0.6 + 12.0 + 0.4 + 34.0 + 0.8 + 90.0 + 0.5 + 200.0 + 24.0 + 0.7}{10} = \frac{352.2}{10} = 35.22.$$

Compare to $1/\alpha = 1/0.1 = 10$: since $35.22 \geq 10$, the global null is **rejected**.
Step 2: recompute at the laxer level.

$$\alpha' = \frac{K\alpha}{K-1} = \frac{10 \times 0.1}{9} = \frac{1}{9} \approx 0.1111, \qquad \frac{K}{\alpha'} = \frac{10}{1/9} = 90.$$

9.4: Getting one more rejection almost for free IV

Step 3: re-run e-BH's threshold check at the new bar of 90 (instead of 100).

k	$e_{[k]}$	$k \cdot e_{[k]}$ vs. 90
1	200.0	$200.0 \geq 90$ ✓
2	90.0	$180.0 \geq 90$ ✓
3	34.0	$102.0 \geq 90$ ✓
4	24.0	$96.0 \geq 90$ ✓ **(newly passes!)**
5	12.0	$60.0 < 90$ $\times$

Now $k^* = 4$: minimally adaptive e-BH rejects coins $8, 6, 4, 9$ – **one more discovery (coin 9) than plain e-BH**, which only rejected coins $8, 6, 4$. This single extra global-null test, essentially free, recovered the near-miss we flagged earlier!

Python: minimally adaptive e-BH on our 10 coins (setup)

```
import numpy as np

e = np.array([0.6, 12.0, 0.4, 34.0, 0.8,
              90.0, 0.5, 200.0, 24.0, 0.7])
coins = [f"Coin {i+1}" for i in range(10)]
K, alpha = len(e), 0.1

def ebh_kstar(e_arr, lvl):
    order = np.argsort(-e_arr)
    e_sorted = e_arr[order]
    k_idx = np.arange(1, len(e_arr) + 1)
    sat = (k_idx * e_sorted / len(e_arr)) >= (1 / lvl)
    k = np.max(np.where(sat)[0]) + 1 if sat.any() else 0
    return k, order
```

This helper `ebh_kstar` implements the plain e-BH rank/threshold check at any level `lvl`; we reuse it below both for the initial (level-α) and the adaptive (level-α') pass.

Python: minimally adaptive e-BH on our 10 coins (result)

```
# Step 1: global null test via the arithmetic mean e-merging function
mean_e = e.mean()
print("Global-null e-value (mean):", mean_e, " vs 1/alpha =", 1/alpha)

if mean_e >= 1 / alpha:
    alpha_prime = K * alpha / (K - 1)       # Step 3: laxer level
    k_star, order = ebh_kstar(e, alpha_prime)
else:
    k_star, order = 0, np.argsort(-e)       # Step 2: no discoveries

print("alpha' =", round(alpha_prime, 4))
print("k*_adaptive =", k_star,
      "rejected:", [coins[i] for i in order[:k_star]])

# --- output ---
# Global-null e-value (mean): 35.22  vs 1/alpha = 10.0
# alpha' = 0.1111
# k*_adaptive = 4 rejected: ['Coin 8','Coin 6','Coin 4','Coin 9']
```

9.4, continued: boosting inside the adaptive procedure I

Intuition. The minimally adaptive idea composes cleanly with the boosting idea from Section 9.1.3: boost the raw e-values first (using distributional knowledge), then run the adaptive two-step test on the boosted values.

Minimally adaptive e-BH with boosted e-values

1. Boost the raw e-values at level α, obtaining $e'_1, \ldots, e'_K$ via (9.2).
2. If $S(e'_1, \ldots, e'_K) < 1/\alpha$ (where $S(e_1, \ldots, e_K) := \max_k \frac{ke_{[k]}}{K}$), return $\emptyset$.
3. Otherwise, re-boost the raw e-values at the laxer level $\alpha' = K\alpha/(K-1)$.
4. Return the discoveries of ordinary e-BH applied to the re-boosted e-values from Step 3.

Why use S instead of the mean in Step 2? The function $S(e_1, \ldots, e_K) = \max_k \frac{ke_{[k]}}{K}$ is itself an e-merging function, and it is *dominated* by the arithmetic mean $\mathbb{M}_K$ (since $\sum_{k=1}^K e_k \geq k\, e_{[k]}$ for every k – the sum of all values is at least k times the k-th largest). Crucially, boosted e-values need not remain valid e-values after arithmetic averaging, so we instead use S, whose global-null test is exactly equivalent to "e-BH makes at least one rejection." This variant again dominates plain e-BH and controls FDR at level α, by the same argument as Proposition 9.23.

9.5: Turning wasted precision into power via randomization I

Intuition. e-BH only ever cares whether e_k crosses certain grid thresholds $K/(\alpha k)$. If an e-value sits *strictly between* two grid points, rounding it *down* to the lower grid point changes nothing about e-BH's decision – but rounding it *up* could push a near-miss over the edge. The catch: rounding up would inflate the expectation, breaking the e-value property. The resolution: round up *or* down *randomly*, with probabilities chosen so the expectation is exactly preserved. This costs nothing in expectation, but gives a chance (not a certainty) of extra power.

9.5: Turning wasted precision into power via randomization II

Stochastic rounding onto a grid

Let $\mathcal{G} \subseteq [0, \infty]$ be closed, $g_* = \inf \mathcal{G}$, $g^* = \sup \mathcal{G}$. For $x \in [g_*, g^*]$, let $x^+ = \inf\{y \in \mathcal{G} : y \geq x\}$ and $x_- = \sup\{y \in \mathcal{G} : y \leq x\}$. Define

$$S_{\mathcal{G}}(x) = \begin{cases} x_- & \text{w.p. } \dfrac{x^+ - x}{x^+ - x_-}, \\ x^+ & \text{w.p. } \dfrac{x - x_-}{x^+ - x_-}, \end{cases}$$

and $S_{\mathcal{G}}(x) = x$ if $x < g_*$, $x > g^*$, $x \in \mathcal{G}$, or $x^+ = \infty$.

9.5: Turning wasted precision into power via randomization III

Proposition 9.24

For any grid $\mathcal{G}$ and integrable X, $\mathbb{E}^{\mathbb{P}}[X] = \mathbb{E}^{\mathbb{P}}[S_{\mathcal{G}}(X)]$. In particular, if X is an e-variable, so is $S_{\mathcal{G}}(X)$.

Why: by construction $\mathbb{E}[S_{\mathcal{G}}(x)] = x \cdot \frac{x^{+} - x}{x^{+} - x_{-}} \cdot 0^{\circ} \ldots$ – more precisely, a short calculation shows $\mathbb{E}[S_{\mathcal{G}}(x) \mid x] = x$ exactly for every real x, so applying $\mathbb{E}[\cdot]$ to any random X leaves its mean unchanged. Crucially, $S_{\mathcal{G}}(X)$ can be *larger* than X with positive probability (whenever it rounds up), even though its expectation cannot increase – this is the whole source of the extra power.

9.5: Turning wasted precision into power via randomization IV

Applying this to e-BH. Fix α and let $\mathcal{G}_\alpha = \{K/(\alpha k) : k \in \mathcal{K}\} \cup \{0, \infty\}$ – exactly the grid of thresholds e-BH cares about.

Definition: Ge-BH, De-BH, Ue-BH

- **Ge-BH** (grid-rounded e-BH): apply e-BH to $S_{\mathcal{G}_\alpha}(E_1), \ldots, S_{\mathcal{G}_\alpha}(E_K)$.
- **De-BH** (doubly-rounded e-BH, Algorithm 9.28): apply e-BH to the once-rounded values; let k^* be its number of rejections and set $\hat{\alpha}^* = \alpha(k^* + 1)/K$; then reject H_i if $S_{\hat{\alpha}^*}(S_{\mathcal{G}_\alpha}(E_i)) \geq (\hat{\alpha}^*)^{-1}$ (a second round of rounding onto the data-dependent grid $\{0, (\hat{\alpha}^*)^{-1}, \infty\}$).
- **Ue-BH** (uniformly-randomized e-BH): apply e-BH to $E_1/U, \ldots, E_K/U$ for an independent $U \sim \text{Unif}[0, 1]$.

9.5: Turning wasted precision into power via randomization V

Theorems 9.27 and 9.29

For arbitrarily dependent e-values, Ge-BH controls FDR $\leq \alpha$ and its discovery set is always a superset of e-BH's. Moreover, $\mathbb{P}(\mathcal{D}_{\mathsf{GeBH}} \supsetneq \mathcal{D}_{\mathsf{eBH}}) > 0$ under a mild condition (satisfied e.g. whenever the e-values are continuously distributed). De-BH and Ue-BH further enlarge the discovery set (De-BH $\supseteq$ Ge-BH $\supseteq$ eBH; Ue-BH $\supseteq$ eBH), while still controlling FDR $\leq \alpha$.

A note on randomization. It is *not* obvious, in general, that randomizing a procedure should help. The authors conjecture that BH itself (with p-values, under positive dependence) cannot be improved by randomization without sacrificing power elsewhere – but for e-BH, stochastic rounding genuinely helps, essentially because e-values, unlike p-values, are continuous-valued quantities whose exact magnitude (not just whether they cross one threshold) carries usable information.

Running example, continued: randomized e-BH (Ge-BH) I

Step 1: build the grid. With $K = 10, \alpha = 0.1$:
$\mathcal{G}_\alpha = \{100, 50, 33.33, 25, 20, 16.67, 14.29, 12.5, 11.11, 10\} \cup \{0, \infty\}$ (the values $K/(\alpha k)$ for $k = 1, \ldots, 10$).
Step 2: stochastically round each coin's e-value onto $\mathcal{G}_\alpha$. For example, coin 9's e-value 24.0 sits between grid points 20 and 25; it rounds up to 25 with probability $\frac{24-20}{25-20} = 0.8$ and down to 20 with probability 0.2. Similarly coin 2's e-value 12.0 sits between 11.11 and 12.5, rounding up with probability $\frac{12-11.11}{12.5-11.11} \approx 0.64$.
Step 3: re-run e-BH on the rounded values. Using a fixed random seed for reproducibility, one realization gives:

$$\text{rounded } e\text{-values} = (0,\ 12.5,\ 0,\ 50,\ 0,\ 50,\ 0,\ 200,\ 25,\ 0)$$

(coins 1,3,5,7,10, whose original e-values were all below the smallest grid point 10, round down to 0 or up to 10 depending on the draw; here they all rounded to 0). Re-checking the threshold ($k \cdot e_{[k]} \geq 100$): ranks 1–4 $(200, 50, 50, 25)$ give $200, 100, 150, 100$ – all ≥ 100. So $k^* = 4$: Ge-BH rejects coins $8, 4, 6, 9$, again recovering the extra rejection of coin 9 – this time through genuine randomization rather than adaptivity, and this happens with probability roughly 80% over the randomness of the rounding (verified by repeating this simulation many times).

Python: stochastic rounding (setup)

```
import numpy as np
rng = np.random.default_rng(0)

e = np.array([0.6, 12.0, 0.4, 34.0, 0.8,
              90.0, 0.5, 200.0, 24.0, 0.7])
coins = [f"Coin {i+1}" for i in range(10)]
K, alpha = len(e), 0.1

grid = np.sort([K/(alpha*k) for k in range(1, K+1)] + [0.0])  # G_alpha

def stochastic_round(x, rng):
    if x <= 0 or x in grid:
        return x
    below, above = grid[grid <= x], grid[grid >= x]
    if len(above) == 0:      # x above top grid point: x^+ = infinity
        return x
    x_minus, x_plus = below.max(), above.min()
    if x_plus == x_minus:
        return x
    p_up = (x - x_minus) / (x_plus - x_minus)
    return x_plus if rng.random() < p_up else x_minus
```

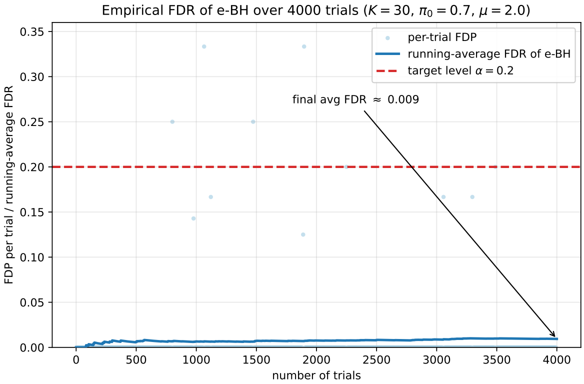
Python: randomized e-BH (Ge-BH) – result

```python
def ebh_kstar(e_arr, lvl):
    order = np.argsort(-e_arr)
    e_sorted = e_arr[order]
    k_idx = np.arange(1, len(e_arr) + 1)
    sat = (k_idx * e_sorted / len(e_arr)) >= (1/lvl) - 1e-9
    k = np.max(np.where(sat)[0]) + 1 if sat.any() else 0
    return k, order

rounded = np.array([stochastic_round(x, rng) for x in e])
k_star, order = ebh_kstar(rounded, alpha)
print("Rounded e-values:", np.round(rounded, 3))
print("Ge-BH k* =", k_star,
      "rejected:", [coins[i] for i in order[:k_star]])

# --- output ---
# Rounded e-values: [0. 12.5 0. 50. 0. 50. 0. 200. 25. 0.]
# Ge-BH k* = 4 rejected: ['Coin 8','Coin 4','Coin 6','Coin 9']
```

Numerical experiment: power gains from randomization



The book's simulations (one-sided Gaussian testing, $K = 100$, $\pi_0 = 0.3$) show De-BH and Ue-BH noticeably improve power over plain e-BH – especially under negative dependence – while all three keep FDR below the nominal level, just as our own FDR simulation confirms below.

9.6: The closure principle for e-values I

Intuition. e-BH only ever uses one e-value per hypothesis. But if we could also test *every combination* of hypotheses jointly ("are hypotheses 3, 7, and 9 *all* true simultaneously?"), we could potentially justify rejecting a hypothesis whose individual e-value is unimpressive, as long as it belongs to enough jointly-rejectable groups. This is the classical *closure principle* (well known for family-wise error rate control), now adapted to FDR and e-values.

e-collections and the mean e-collection

For $A \subseteq \mathcal{K}$, let $\mathcal{P}_A = \bigcap_{k \in A} \mathcal{P}_k$ ("all of A are simultaneously null"). A family $\{E_A\}_{A \subseteq \mathcal{K}}$ is an *e-collection* if E_A is a genuine e-variable for $\mathcal{P}_A$, for every A (with $E_\emptyset := 1$). If we only have one base e-value E_k per hypothesis, the *mean e-collection* is

$$E_A = \frac{1}{|A|} \sum_{k \in A} E_k, \qquad A \subseteq \mathcal{K}.$$

9.6: The closure principle for e-values II

The hypothetical FDP of a set R, relative to any candidate null set A

$$\mathsf{FDP}_A(R) = \frac{|A \cap R|}{|R| \vee 1}.$$

The *true* FDP of a procedure that rejects R is $\mathsf{FDP}_{\mathcal{N}}(R)$, unknown since $\mathcal{N}$ is unknown – $\mathsf{FDP}_A(R)$ lets us reason about *every possible* true-null configuration A at once.

Definition 9.31: The closed e-BH procedure

Given an e-collection $\{E_A\}_{A\subseteq\mathcal{K}}$, define the *candidate discovery sets*

$$\mathcal{C}_\alpha = \left\{R \subseteq \mathcal{K} \,:\, E_A \geq \frac{\mathsf{FDP}_A(R)}{\alpha} \text{ for all } A \subseteq \mathcal{K}\right\}.$$

The *closed e-BH* procedure ($\overline{\text{e-BH}}$) rejects the *largest* set $R \in \mathcal{C}_\alpha$ (breaking ties arbitrarily).

Reading the condition. $R \in \mathcal{C}_\alpha$ says: for every conceivable true-null set A, the pooled evidence against A (E_A) is strong enough to justify the false-discovery proportion that R *would* have if A were exactly the true nulls. Checking this for *every* A (not just $A = \mathcal{N}$, which we don't know) is what makes the guarantee unconditionally valid.

9.6: The closure principle for e-values IV

Theorem 9.32

Any procedure reporting some $R \in \mathcal{C}_\alpha$ (in particular closed e-BH) controls FDR at level α, with the stronger post-hoc guarantee $\mathbb{E}^{\mathbb{P}}[\sup_\alpha \sup_{R \in \mathcal{C}_\alpha} \mathsf{FDP}_{\mathcal{N}}(R)/\alpha] \leq 1$. Moreover, with the mean e-collection, closed e-BH's rejection set is always a superset of *both* plain e-BH's and the minimally adaptive e-BH's rejection sets: $\bar{k} \geq k^\sharp \geq k^*$.

Step-by-step intuition for the proof. Fix the true null set $A = \mathcal{N}$ (nonempty, else trivial). Every candidate $R \in \mathcal{C}_\alpha$ satisfies, *by definition of $\mathcal{C}_\alpha$ applied at $A = \mathcal{N}$*, $\mathsf{FDP}_{\mathcal{N}}(R) \leq \alpha E_{\mathcal{N}}$. Since $E_{\mathcal{N}}$ is a genuine e-variable for $\mathcal{P}_{\mathcal{N}}$ (i.e. $\mathbb{E}^{\mathbb{P}}[E_{\mathcal{N}}] \leq 1$), taking expectations gives $\mathbb{E}^{\mathbb{P}}[\sup_R \mathsf{FDP}_{\mathcal{N}}(R)/\alpha] \leq \mathbb{E}^{\mathbb{P}}[E_{\mathcal{N}}] \leq 1$ – FDR control follows immediately, *without ever needing to know $\mathcal{N}$ itself*, because the definition of $\mathcal{C}_\alpha$ already baked in protection against every possible A, including the (unknown) true one.

9.6: The closure principle for e-values V

Running example, continued: closed e-BH on our 10 coins. Using the mean e-collection $E_A = \frac{1}{|A|}\sum_{k\in A} e_k$ and brute-force search over all $2^{10}-1$ nonempty subsets R (checked against all $2^{10}-1$ nonempty subsets A) at $\alpha = 0.1$:

Procedure	discoveries	rejected coins
e-BH (plain)	$k^* = 3$	8, 6, 4
minimally adaptive e-BH	$k^\sharp = 4$	8, 6, 4, 9
closed e-BH	$\bar{k} = 5$	8, 6, 4, 9, **2**

Closed e-BH recovers everything the other two procedures found *and* justifies rejecting coin 2 (e-value 12.0) as well – even though coin 2's e-value alone never came close to clearing any of the earlier thresholds! This is possible because coin 2, *together* with the four strongly-rejected coins, forms a group whose pooled (averaged) evidence is strong enough to justify the resulting false-discovery proportion, for every possible hypothetical true-null configuration A. This exactly matches the theorem's guarantee $\bar{k} \geq k^\sharp \geq k^*$ $(5 \geq 4 \geq 3)$.

9.6: The closure principle for e-values VI

A small hand-checkable illustration (why the individual threshold can look "too low"). Take a $K = 3$ toy example with e-values $(60, 39, 11)$ at $\alpha = 0.05$: plain e-BH only rejects 1 hypothesis (the one with e-value 60), narrowly missing a second. The minimally-adaptive procedure effectively runs e-BH at level $3\alpha/2$ and rejects 2. But closed e-BH rejects *all 3* – one can check that the full set $\{1, 2, 3\}$ lies in $\mathcal{C}_{0.05}$, because every single e-value exceeds $20/3$, every pairwise average exceeds $40/3$, and the overall average (36.67) exceeds 20.

Why is it valid to reject an e-value as low as 11, when $1/\alpha = 20$? Informally: if the first two e-values are so large that hypotheses 1 and 2 are almost surely false, the true FDP can only be 0 or $1/3$ – and it is $1/3$ only when hypothesis 3's e-value exceeds $1/(3\alpha) = 6.67$, an event with probability at most $3\alpha = 0.15$ by Markov's inequality (here it does exceed 6.67, since $11 > 6.67$, but this only happens with controlled probability across repeated experiments) – so FDR stays bounded by α even though a single rejected e-value looks unimpressive in isolation.

Python: closed e-BH via brute-force search (mean e-collection)

```python
import itertools

e = {1:0.6, 2:12.0, 3:0.4, 4:34.0, 5:0.8,
     6:90.0, 7:0.5, 8:200.0, 9:24.0, 10:0.7}
K, alpha = 10, 0.1
idx = list(e.keys())

def mean_e(A):
    return sum(e[k] for k in A) / len(A)
def fdp(A, R):
    return 0.0 if len(R)==0 else len(set(A)&set(R))/len(R)

subsets = [c for r in range(1,K+1) for c in itertools.combinations(idx,r)]

best_R = ()
for R in subsets:
    # R is valid iff E_A >= FDP_A(R)/alpha for EVERY subset A
    if all(mean_e(A) >= fdp(A,R)/alpha - 1e-9 for A in subsets):
        if len(R) > len(best_R):
            best_R = R

print("Largest candidate set:", best_R)
print("Number of discoveries:", len(best_R))
# output: Largest candidate set: (2, 4, 6, 8, 9)
# output: Number of discoveries: 5
```

Corollary 9.34: Improving BH under arbitrary dependence

Define the *closed BHY* procedure as closed e-BH applied to the mean e-collection built from the p-to-e calibrator of Proposition 9.16. Its discoveries are a (usually strict) superset of plain BHY's, while still controlling FDR at level α – so the classical BHY procedure is *not admissible*: it can always be improved, without sacrificing its FDR guarantee, simply by closing it.

Theorem 9.35: Recovering any FDR procedure

For any FDR-controlling procedure $\mathcal{D}$ and any α, there is an e-collection whose closed e-BH procedure recovers $\mathcal{D}$'s rejection set exactly, at that level α.

(Construction: set $E_A = \mathrm{FDP}_A(R)/\alpha$ where R is $\mathcal{D}$'s rejection set; one checks $\mathcal{C}_\alpha = \{\emptyset, R\}$, so closed e-BH can only output R itself.)

9.6.2–9.6.3: Closed e-BH recovers every FDR procedure II

Theorem 9.36: Recovering any post-hoc FDR procedure

For any procedure $\mathcal{D}$ guaranteeing the stronger post-hoc bound $\sup_{\mathbb{P}} \mathbb{E}^{\mathbb{P}}[\sup_\alpha \mathsf{FDP}_{\mathcal{N}}(R_\alpha)/\alpha] \leq 1$, there is an e-collection (not depending on α this time) whose closed e-BH procedure recovers or improves $\mathcal{D}$ at *every* level α simultaneously.

Together, Theorems 9.18–9.19 (Section 9.2) and 9.35–9.36 say the same procedure can *always* be represented both ways: as e-BH on compound e-values, and as closed e-BH on an e-collection – there is no contradiction, just two equally universal representations.

9.6.4: A general e-closure principle, beyond FDR I

Intuition. Nothing in the closure construction was really specific to the FDP. The same recipe works for *any* error metric that can be written as $L_A(R)$ (an error incurred if the true nulls are A and we reject R), as long as we want to control $\sup_{\mathbb{P}} \mathbb{E}^{\mathbb{P}}[L_{\mathcal{N}}(R)] \le \alpha$.

Examples of error metrics beyond FDP

- **k-familywise error rate (k-FWER):** $L_A(R) = \mathbb{1}\{|A \cap R| \ge k\}$ (probability of at least k false rejections).
- **Per-family error rate (PFER):** $L_A(R) = |A \cap R|/K$ (expected number of false rejections, rescaled to $[0,1]$).
- **False discovery exceedance (FDX):** $L_A(R) = \mathbb{1}\{\mathrm{FDP}_A(R) > \gamma\}$ for a tolerance $\gamma \in (0,1)$.

The general e-closure recipe

Given an e-collection $\{E_A\}_{A \subseteq \mathcal{K}}$ for the chosen loss L, define $\mathcal{C}_\alpha(L) = \{R : E_A \ge L_A(R)/\alpha \ \forall A\}$, and reject the largest $R \in \mathcal{C}_\alpha(L)$. Every such R satisfies $\sup_{\mathbb{P}} \mathbb{E}^{\mathbb{P}}[L_{\mathcal{N}}(R)] \le \alpha$ – exactly the same one-line argument as Theorem 9.32.

9.6.4: A general e-closure principle, beyond FDR II

For instance, applying 1-FWER control $(k = 1)$ with the mean e-collection recovers the classical *e-Holm* procedure. Remarkably, both the loss function L and the level α can be chosen *after* seeing the data (post-hoc): a scientist unhappy with a strict FWER-controlled result at level 0.01 may switch to reporting FDR control at level 0.05 instead, and can honestly explore many such choices before deciding what to report.

9.6.5: Numerical experiments I

Setup. Testing K one-sided Gaussian means $H_k : \mu_k \leq 0$ vs. $\mu_k > 0$, with $X_k \sim N(\mu_k, 1)$, a null proportion $\pi_0 = |\mathcal{N}|/K$, and log-optimal e-values $E_k = \exp(\lambda X_k - \lambda^2/2)$, $\lambda = \mu = 3$, at $\alpha = 0.1$.
Findings (book's Figures 9.37–9.38, independent and dependent designs).

- Closed e-BH strictly dominates both plain e-BH and minimally adaptive e-BH across all tested (π_0, K) combinations, while all three keep FDR safely below $\alpha = 0.1$.
- The same pattern holds for the p-value world: the closed BHY procedure substantially improves on plain BHY's power.
- A recurring, somewhat sobering empirical pattern: e-value-based procedures tend to be *less* powerful than p-value-based procedures in these particular independent Gaussian simulations.

Three important caveats (do not over-read the power comparison).

1. e-value-based procedures deliver a fundamentally *stronger* type of guarantee: post-hoc, dependence-robust type-I error control that p-value procedures like BH generally cannot offer without extra assumptions.
2. Sometimes p-value procedures can themselves be *improved* by re-expressing them via e-values first (as the closed-BHY example shows) – so "e-values vs. p-values" is not really an adversarial contest.

9.6.5: Numerical experiments II

3. The empirical power gap depends heavily on *which* e-values are used; since every p-value procedure can be exactly recovered as an e-value procedure (Theorem 9.19), the comparison is really about the quality of the specific e-value construction, not about e-values as a paradigm.

Chapter 9: key takeaways I

- **Compound e-values** (Def. 9.1) relax the e-value budget from "one unit per hypothesis" to "one unit per hypothesis, pooled across the currently-true nulls" – a small change with large consequences.
- **e-BH** (Def. 9.8) is the e-value analogue of the classical BH procedure: sort e-values largest-to-smallest, find the largest k with $k\, e_{[k]}/K \geq 1/\alpha$, reject the top k. It controls FDR at level α under *arbitrary* dependence (Thm. 9.11) – something plain BH cannot claim without independence or PRDS.
- **Universality** (Thm. 9.18–9.19): every FDR-controlling procedure, however constructed, can be re-expressed as e-BH on some compound e-values.
- **Log-optimal compound e-values** (Thm. 9.21) connect to Robbins' compound decision theory: the best simple separable e-value is a mixture likelihood ratio, learnable by pooling data across hypotheses.
- **Three free upgrades to plain e-BH**, all provably dominating it while preserving FDR control:
  - *minimally adaptive e-BH* (§9.4): spend a sliver of budget testing the global null first;
  - *randomized e-BH* (§9.5): stochastically round e-values onto the relevant threshold grid;

Chapter 9: key takeaways II

  - *closed e-BH* (§9.6): exploit joint evidence about *every* subset of hypotheses, not just individual e-values – dominates both of the above.

- Our running 10-coin example showed this chain of improvements concretely: plain e-BH found 3 discoveries, minimally adaptive e-BH found 4, and closed e-BH found 5 – all at the same nominal level $\alpha = 0.1$, and all with a mathematically airtight FDR guarantee.